Voice and its Audio-Contents

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Shifts in technology bring with them new configurations of embodiment, and in addition, resituate how voicing comes to make incarnate a sense of self.
—Brandon LaBelle, 2010

Introduction

In 1946-7, Dennis Gabor proposed his 'Theory of Communication'[1] and 'Acoustical Quanta and the Theory of Hearing'[2]. Unlike Fourier Analysis, which established a dialectic between time and frequency domains of representation for a signal, Gabor sought unification of the two as a time-frequency localization. While Fourier analysis treats signal information as an infinite sum of frequency components, Gabor saw signal information as composed of rectangular time-frequency cells which he calls quanta. Gabor's configuration of information can be visualized as spectographs or *'sound portraits… which contain most of its subjectively important features'*[3] This theory of information also establishes an uncertainty relationship in measuring both the time and frequency content of a quanta of information which resonates with perceptual limits of the ear in the acoustic model of hearing. This model of communication and information has had a vast impact on current day signal processing applications for speech detection and image recognition.

As accounted by Curtis Roads' in his Microsounds' text, Gabor's ideas were especially influential to Norbert Wiener's 1964 book on *Spatio-Temporal Continuity, Quantum Theory and Music*[4] as well as for Iannis Xenakis' 1971 book on *Formalized Music*[5]. Roads as a major contributor to the genre of musique concrète through granular synthesis. In 1981, Curtis Roads was able to rework Xenakis's method by implementing granular synthesis on a digital computer, and remained committed to continual experimentation and development of granular synthesis techniques through his career. Reminiscent of Kandinsky's *Point and Line to Plane*, Roads issued a composition in 2003 called Point Line Cloud. In this work, he demonstrates the variety of time-scales through which a sound composer can work. Especially interesting in his compositional style is the development of hardware and software to experiment with mathematical ideas and chart out techniques in sound synthesis. These techniques share much in the tradition of impressionist and pointillist painters in which sound is treated as a discrete point, and through combination with others, may form higher level artistic structures.

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[1] Gabor, "Theory of Communication. Part 2."
[2] Gabor, "Acoustical Quanta and the Theory of Hearing."
[3] Gabor.
[4] Wiener, "Spatio-Temporal Continuity, Quantum Theory and Music."
[5] Iannis Xenakis, *Formalized Music Thought And Mathematics In Composition*.

While there is a direct connection from Gabor to Xenakis and Roads through sonic expression of mathematical theories of information and hearing, there also exists a connection between advancements in audio technologies and their applied use in psychotherapy. Starting in America in the 1930s, psychoanalysts Harold Laswell and Earl F. Zinn began implementing audio inscription technologies into their practices with patients. They developed a technique referred to by Michael Lampert as Fine-Grained Analysis through which a patient's voice and therapeutic session were recorded on wax cylinders, subject to rigorous and lengthy group analysis, and characterized for minute irregularities. This increased value in the recorded voice as an analytical tool and representation of the patient is a direct effect of the newfound power to inscribe speech into a material and playback at any time. Further, it allowed listeners to target and analyze precise moments within an audio recording by applying a microanalytical window after a talk therapy session. This reconfiguration of the patient-psychologist dynamic alerts a growing approval in the confidence of utilizing microsound analysis to objectively analyze or understand the patient.

In this essay, it will be contended that the relation between patient and therapist including voice through grain analysis has and will continue to exist within the technological arrangements made possible by handheld smart phones equipped with digital microphones, memory, and machine-learning based software frameworks fueled by digital signal characteristics. This analysis will investigate the case of Sonde Health in their mental fitness recognition smartphone app. In this process it will be revealed that a desire to objectify the unconscious is common between the tedious efforts of Laswell and Zinn and the emergence of sound analysis software based on machine learning of signal processing data. Through the consideration of Sonde, an alternative relation will be speculated which invites the speaker or patient to hear, assess, and reconstruct the grains of their voice anew. The ingredients for this method will be built in considering William Burroughs cut up technique as a process of integrating randomness, collage, and micro-montage into the process of multimedia expression which resists device operativity. Secondly, the UPIC system developed by Iannis Xenakis will provide a framework for the connection of human gesture to production of sound. Lastly, Jonathan Sterne's description of volutes in his Diminished Faculties will help explore voice as an external phenomena.

Harold Laswell and Earl Zinn

In 'Fine Grained Analysis' Lampert provides a historical account of the emerging use of recording technologies for psychotherapy. Psychoanalytic tradition of the early 20th century was guided by Freud's talk therapy method which *'advised the analyst to let attention float freely and avoid note taking'* as '*He couldn't stop and write, and most felt that notes written afterward would be selective owing to the limitations of memory. Nor could you recruit a note taker, because of pernicious observer effect.'*[6] This observer effect indicates that if anyone else is present during the psychoanalytic session, such as a note taker, conditions for free association will be broken. Secondly, the analyst is viewed as an imperfect observer and the act of writing

[6] Lempert, "Fine-Grained Analysis."

notes during or after the session may be subject to selection bias. In response to these problems, Zinn and Laswell sought out electro-mechanical recording apparatuses compatible with the free association process to provide an immediate inscription of the therapeutic conversation for later review and analysis.

In 1929, Earl Zinn led a research initiative for the National Research Council on a committee for the study of problematic interaction of personalities within families. Through private funding, Zinn was able to begin development and experimentation of electro-mechanical recording apparatuses for use in psychotherapy. Zinn partnered with Alexander Graham Bell's Dictaphone Company and was able to make the apparatus minimally invasive by placing it out of sight *in an adjoining room to avoid feedback but especially to hide the equipment and mute its noise. He used a condenser microphone and a four-stage amplifier, and his customized dictation machine featured a double mandrel for continuous recording*'[7] Patient-analyst sessions were recorded on wax cylinders, which due to their novelty and expense at the time, needed to be shaved and reused for each session. In these experiments, transcribers were hired to accurately dictate the content of what was said during the session. Lasswell at the University of Chicago was inspired by Zinn's experiments; however, he sought more than just the transcribed conversation of a session. Lasswell believed that somatic indicators of the body can serve as representations of the unconscious state at times where speech may betray the speaking subject. Lasswell was interested in *'pinpointing embodied indexes of interiority such as galvanic skin response, heart rate, and respiration'.* Additionally, He '*would jot down "movements of the subject," and, to synchronize these, "a time marker [would be] ... used to make a signal on the moving film and in the acoustic machinery*" Shifting from the tradition of conversation analysis, 'His research epitomized a "subjective turn' in the social sciences in which one strained to spy the mind through the apperature of the body"[8]

In continuing this method, Zinn further abstracted the analysis of conversation to include speech specific characteristics. He *'transcribed variations in speech delivery such as false starts, cut-off speech, and repetitions*'[9] This ability to pinpoint such characteristics of the speaking subject were only possible through mechanical recording and playback technologies. This drive towards finer grained inscription and analysis of the subject's voice, is paired to an implicit desire for complete objectification of the communicative unconscious. While the traditional Freudian talk-therapy approach was guided by the idea that the analyst could reach connection to the patient's unconscious through free-association, the functionality of sound recording technologies shifted psychoanalysis towards creation of an objective inscription of the patient's voice which serves as a material representation of the self.

Just as the analyst was viewed as a limited objective participant in traditional Freudian psychoanalysis, the tedious indexing of fine-grained analysis came to its own limits. To characterize the voice in such a detailed time precision required a huge amount of manpower. Lambert includes that *"About 120 hours were required to transcribe phonetically the first*

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[7] Lempert.
[8] Lempert.
[9] Lempert.

half-hour of the interview," he reported, and another "20 hours to retranscribe a fifth of this material and analyze it functionally."[10] To ensure feasibility, analysts began to hone in on shorter segments of therapeutic conversation to limit the amount of time needed to transcribe and perform analysis. One such move was to only look at the first five minutes of a voice sample. It was believed that 5 minutes was enough to provide a representative sample of the self from which the analyst could make inferences. Second, the shift in agency of mechanic recording technologies in psychoanalysis altered the training and techniques of analysts. Lambert mentions that *"in the teaching of psychotherapy ...it is important that every inflection of the voice, every whisper, yawn, sigh, slight and almost imperceptible dropping or raising of the voice be recorded with lifelike quality."* Further, "By acting, in effect, like the fine-grained analyst, the clinician could cultivate indexical receptivity. Recording-based analysis was not only an epistemological prosthesis but a pedagogy for the clinical sensorium."[11]

As recording technologies became more widely available and inexpensive via cassette tape, the playback therapeutic technique was born. Levy-Landesberg and Pinchevski describe the tape recorder as breaking down the analyst's traditional *"role of bouncing back the patient's voice (Dolar, 2006), in therapeutic playback the interval between patient-as-speaker and patient-as-listener is relegated to the interval between pressing the 'record' and 'play' buttons."*[12] As opposed to Lasswell and Zinn's experiments with micro-analysis on wax cylinders, this method permitted the patient to re-view and re-listen to their voice outside of the therapeutic setting. This was particularly influential in the Exposure and Response-Prevention (ERP) method which aimed to treat Obsessive Compulsive Disorders through repetitive hearing of one's own compulsive thoughts or fears. Rather than viewing the recorded voice as a material to be transcribed and decoded by the analyst, it is playing an active role as a sub-vocalized stimulation for the patient. In other words, 'ERP collapses past-self and present-self, producing instead a competing and repetitive… stream of consciousness' such that 'the patient's own voice is remediated through tape recording in an attempt to suture it back to its source.'[13] In this method, the therapeutic transference relation shifts from analyst-patient to tape-recorder-patient as the patient enters into a feedback loop with their own disembodied voice. As will be discussed in the next section, this rearrangement of the psychoanalytic practice by inclusion of recording technologies for objective fine-grained analysis has ingredients which continue to live on within modern signal processing based approaches toward diagnostics of mental state.

Sonde Health

Jumping forward several decades and media platforms, a new genre of technology has emerged called vocal biomarking. To preface, vocal biomarking is not a therapy; It uses analytical characteristics of a voice signal to detect illnesses in a speaking voice. In many applications, vocal biomarking resembles a traditional medical tool such as blood sampling for

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[10] Lempert.
[11] Lempert.
[12] Levy-Landesberg and Pinchevski, "The Recording Cure."
[13] Levy-Landesberg and Pinchevski.

diagnosis of cardiovascular or respiratory illness. A vocal biomarking company called Sonde Health created a smart-phone application based on these principles, however, they aim to diagnose mental health issues. Specifically, they claim objective measurement of anhedonia symptoms such as reduced pleasure or lack of interest in life experiences previously found desirable. Anhedonia can be indicative of several common mood disorders such as depression, anxiety, post-traumatic stress disorder, and bipolar syndrome.

The app's design-concept mimics that of existing physical health monitoring services like in MyFitnessPal, Strava, or Apple Health. In Sonde Health's 2024 study completed at the Cognitive Behavioral Institute in Pittsburgh, PA, they note that *'providing users with this familiar fitness tracking framework could facilitate better understanding and improving of their mental well-being within an integrated digital ecosystem, extending the benefits seen in the physical health domain to empower individuals in actively managing their mental fitness.'* [14]By incorporating the familiar aesthetic and language of physical health software and measurement of objective quantitative characteristics of the voice, Sonde establishes itself strategically within the shifting social climate of mental health or fitness. In fact, their analytical voice software culminates in a daily Mental Fitness Vocal Biomarker Score (MFVB) with ranges of "Excellent (80-100), "Good" (70-79), and "Pay Attention" (0-69). The software does not analyze speech content due to privacy concerns; however, it can index several vocal attributes such as jittering, shimmering, pitch variability, energy variability, vowel spacing, phonation duration, speech rate, and pause duration. The app includes *'a voice journaling capability where users can record 30 second voice samples in response to a range of predefined prompts or talk about their own thoughts and feelings. The response is automatically transcribed and stored in a journaling section'* [15] Integrating the vocal biomarker diagnostic tool into a user-friendly application implies a desire for the metric to serve as a hermeneutic feedback indicator for therapeutic improvement. Sonde expresses: *"The "Mental Fitness" concept underscores the opportunity for individuals to engage in healthy behaviors that can proactively mitigate mental health risks, fostering well-being and mental resilience akin to how physical fitness promotes overall health and vitality"*[16]

Sonde claims, *'No machine learning models were used in any part of the MFVB score algorithm'*[17]*.* A closer look into their patent, *US10548534B2 - System and method for anhedonia measurement using acoustic and contextual cues,* suggests otherwise. In the patent, they write:

> *'by combining the extracted plurality of acoustic features and said contextual data received from said contextual sensors, wherein a combination of Random Forest and Gini Impurity is used for selecting said extracted acoustic features, wherein said selected acoustic features are weighted and entered into predictive models, wherein said*

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[14] Larsen et al., "Validating the Efficacy and Value Proposition of Mental Fitness Vocal Biomarkers in a Psychiatric Population."
[15] Larsen et al.
[16] Larsen et al.
[17] Larsen et al.

predictive models determine an overall score of anhedonia, and wherein said contextual data determines component scores of anhedonia'[18]

Random Forest and Gini Impurity are both supervised machine learning techniques within the subcategory of decision tree learning. The acoustic characteristics analyzed include the sub-categories of time domain, amplitude, spectral, and perceptual features. Time domain features include zero-crossing rate; Amplitude features include waveform minimum and maximum; Spectral features include short-time autocorrelation functions or cepstrum describing periodic structures in frequency; Perceptual features include pitch or resonant frequencies in the vocal tract. It may be desirable for Sonde to distance themselves from classification as a machine-learning based solution due to biases arising from sample data used to train their models. Subsequently, it reinforces the objective nature of their diagnosis as it is centered on measurement and mathematical interpretation of the voice signal.

As a clear analog to fine grained analysis, Sonde transforms the voice into an indexed representation at small time steps through which psychological conclusions can be inferred within the speaking patient; however, in this case it is substantially reduced from 5 minutes to 30 seconds. It expands upon the idea that micro-gestures can serve as analytical points where voice irregularities are recognized through a form of machine hearing which reinterprets voice as signal. The analytical framework is based around a building up of sensorium in the form of a machine-learning model to recognize qualities indicative of anhedonic voice signals. Instead of training analysts to develop a capacity for hearing, recognizing, transcribing, and indexing vocal nuances, a machine learning model can serve as a perfect substitute. This significantly cuts down on the time required to transcribe, analyze, and make a real-time conclusion about the patient. In fact, this method doesn't require any intervention from a psychotherapist and can be done from any location. In many ways, Sonde's application is composed of several technological and conceptual ingredients contained within Fine Grained Analysis and the Playback Technique. Initial desires for an objective unconscious are reframed in *'the importance of speech features in psychiatric disorders as objective, reproducible, and time-efficient biomarkers.'*[19] In the next section, a closer look will be taken to the lineage of signal cepstrum, their applications outside of vocal biomarking, and application in Xenakis's UPIC system.

The UPIC

A defining trait of Sonde's app is its emphasis on calculating acoustical features of an audio sample. In their patent US10548534B2**,** they describe in words and a diagram how *'an acoustic feature (acoustic descriptor) can be segmented from a time series of audio data…'* where *'A typical frame lengths can vary from 10 to 35 ms.'*[20] This process is characteristic of many other

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[18] Chen and Jordan, System and method for anhedonia measurement using acoustic and contextual cues.
[19] Berardi et al., "Relative Importance of Speech and Voice Features in the Classification of Schizophrenia and Depression."
[20] Chen and Jordan, System and method for anhedonia measurement using acoustic and contextual cues.

speech recognition and audio signal processing techniques. As mentioned in the introduction to this paper, Dennis Gabor was among the first to conceptualize the utility in windowing audio signals referred to as Gabor wavelets, grains, or quanta. Gabor originally formed grains by windowing signals with a standard bell-shaped Gaussian envelope. Several windowing techniques have developed since Gabor including John Wilder Tukey's Tukey window as described and implemented by Curtis Roads in his first experiments with granular synthesis. Tukey is mentioned here as an important pioneer in laying the foundations for speech analysis. He is author of the Fast Fourier Transform algorithm (FFT, 1965)[21] and Cepstrum Analysis (1963)[22] among many other contributions to mathematics and signal processing. It is also noted that computer researcher and artist Michael Noll contributed to Cepstrum Pitch Determination (1967)[23]. For Cepstrum Analysis, the audio signal is broken up into small chunks typically on the order of milliseconds, whereby the harmonic frequency content or spectrogram can be extracted. Tukey's method was *'satisfactory for determining the voice pitch of voiced speech, so speech analysis was one of the earliest applications, much of the development being done at Bell Telephone Labs'* [24] Just as Cepstrum Analysis is useful for Sonde Health in their mental fitness app, it has a vast number of applications especially for problems in the field of music information retrieval such as automated music transcription[25] and instrument source separation[26]. We now turn to early explorations of windowed signal techniques in the genre of musique concrète.

As described by Curtis Roads in Microsounds, granular synthesis is:

> *'the granulation of sampled sounds is a powerful means of sound transformation. To granulate means to segment (or window) a sound signal into grains, to possibly modify them in some way, and then to reassemble the grains in a new time order and microrhythm. This might take the form of a continuous stream or of a statistical cloud of sampled grains.'*[27]

As a composer, Roads identifies a spectrum of time-scales for which sound exist. Most relevant among them is the macro-scale which amounts to the duration of a song, the meso-scale which may be smaller groupings or measures in the form of the composition, the sound object which is the single note of traditional music, and the micro-scale which is sound particles at timescales near the threshold of hearing. The focus of Road's experiments is to find techniques of grain assembly at the micro-scale that generate and evolve larger scale sonic forms such as clouds with dynamic densities and textures. Roads identifies Iannis Xenakis as the first granular synthesis composer with his 1959 Analogique B recording which was created *'by recording sine tones on tape, cutting the tapes into hundreds of tiny pieces, and then recombining by manual*

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[21] Cooley and Tukey, "An Algorithm for the Machine Calculation of Complex Fourier Series."
[22] Bogert, Healy, and Tukey, "The Quefrency Analysis of Time Series for Echoes."
[23] Noll, "Cepstrum Pitch Determination."
[24] Randall, "A History of Cepstrum Analysis and Its Application to Mechanical Problems."
[25] Bittner et al., "A Lightweight Instrument-Agnostic Model for Polyphonic Note Transcription and Multipitch Estimation."
[26] Engineering, "Understanding by Unmixing."
[27] Roads, *Microsound*.

splicing according to a stochastically-generated score.'[28] As digital computing became more widely available, granular synthesis programs and software applications were made to assist in composition. Among them is the Xenakis's Unité Polyagogique Informatique du CEMAMu (UPIC) which translates to Computer Musical Composition Tool in english.

The UPIC system included a large interface reminiscent of a painter's easel which enabled composers to sketch sounds. The interface's 2D plane was organized with time spanning the horizontal axis and frequency spanning the vertical axis. Lines drawn onto the interface page are called arcs where a *'page can have 64 simultaneous arcs, with four thousand arcs per page. Most importantly, the duration of each page can last from 6 ms to more than two hours. This temporal flexibility lets the user zoom in to the micro time scale. When a page lasts only a second, say, any arcs written onto it will be microsounds.'*[29] This ability to zoom and draw at multiple time-scales resembles Road's broadened compositional perspective. In his traceological analysis of artworks created using the UPIC system, Mauricio Arturo Meza Ruiz writes: *'When working with the UPIC, the composer is faced with the possibility of designing all the formal and temporal aspects of the musical work, from the waveform to the macrostructure, passing through all the intermediate levels.'* [30] What is an essential design feature of UPIC is the hand through drawing as the intermediate between individual and musical apparatus. This model allows *'the creator to integrate his listening and body movement to a process of creation that implies an immediate return of information by the computer, also allows the recording of this creative, corporeal, and perceptive activity'* [31] The design allows users to engage in a form of poetics which reconfigures *'the relationship between technique, body movement, auditory and visual perception, and imaginary and creative intentionality, within the context of musical creation.'* [32] Ruiz emphasizes that Xenakis cultivated his vision for the tool from 1950 until its implementation in 1977. Xenakis believed that creativity is essential to human individuality and that *'society must make available tools that allow everyone to experiment and amplify their creativity.'* [33] When asked about the intention of including the hand in his design, Xenakis elegantly remarks:

> *'The hand, itself, stands between randomness and calculation. It is both an instrument of the mind - so close to the head - and an imperfect tool. The products of the intelligence are so complex that it is impossible to purify them in order to submit them totally to mathematical laws. Industrialization is a forced purification. But you can always recognize what has been made industrially and what has been made by hand. Industrial means are clean, functional, poor. The hand adds inner richness and charm.'*[34]

While the UPIC system shares roots with Lasswell, Zinn, and Sonde's micro-time treatment of sound as, it also affords opportunities for the sound-creator to pass between a variety of

[28] Roads, "From Grains to Forms."
[29] Roads, *Microsound*.
[30] Meza, "20. An Approach to the Epistemic Potential of the UPIC."
[31] Meza.
[32] Meza.
[33] Meza.
[34] Xenakis, Brown, and Rahn, "Xenakis on Xenakis."

compositional scales. Secondly, while the playback technique emphasizes an artificially induced subvocalization of obsessive and fear provoking thoughts, the UPIC system affords a different feedback loop between gestures of the body through drawing and abstract sounds representing dynamic energy and emotional states. As we will see next, Xenakis was not the only one playing around the cutting of magnetic tapes and randomness in the 2nd half of the 20th century.

The Cut-Up

William Burroughs is credited as founder of what he calls ‘The Cut Up Method’. Burroughs makes it clear that his inspiration for the technique stems from surrealist member Tristan Tzara’s method of poem creation through random selection of words from a hat in 1920. One can also connect this tradition, as did Brandon LaBelle, to the Letterist International and Ultra-Letterist sound poetry movements of the 1930s - 1960s such as Kurt Schwitters *Ursonate (1932)*. While Burroughs fits within a lineage of experimentation with the written medium, he is perhaps the first to apply this method to the creation of long form novels. Similar to Xenakis, Burroughs describes the algorithmic nature of his method:

> *‘The method is simple. Here is one way to do it. Take a page. Like this page. Now cut down the middle and cross the middle. You have four sections: 1 2 3 4 ... one two three four. Now rearrange the sections placing section four with section one and section two with section three. And you have a new page.’*[35]

While Xenakis views the hand as an essential source of randomness in the creation of art, Burroughs sees the scissor as an *‘unpredictable spontaneous factor’*[36] when used to cut up, algorithmically rearrange, and collage sections of found or self-written material. While this method was originally applied to the medium of the written word, Burroughs was quick to experiment with its use in audio tape and film.

In moving from word to magnetic tape, Burrough’s thought process behind the cut-up technique is centered around resistance towards the operativity of the playback device. The written word serves as both an inscription medium and playback medium through words arranged in a linear left right and top down fashion. By geometrically reconfiguring the order for rectangular regions of a text and combining several different texts from a variety of authors, Burroughs resists the culturally inscribed operations of reading and writing. Likewise with magnetic tape, Burroughs may choose to cut the tape vertically, horizontally or in a variety of geometries and suture back together to be played back and heard. Katherine Hayles describes this approach as:

> *‘The tape-recorder acts both as a metaphor for these mutations and as the instrumentality that brings them about. The taped body can separate at the vertical "divide line," grotesquely becoming half one person and half another, as if it were tape spliced lengthwise. In a disturbingly literal sense, the tape-recorder becomes a*

[35] Baraka, *The Moderns*.
[36] Baraka.

> *two-edged sword, cutting through bodies as well as through the programs that control and discipline them.'*[37]

The electronic circuits of tape-recorders were designed to encode sound and speech into a magnetic tape inscription which also bears a decoding program to recover and make audible again (re-play) the tape. By cutting the tape vertically and reassembling it out of order, he resists the fixed linear time dimensionality of speaking and listening. By combining fragments of tape from multiple voices, he resists the imposition that a tape should represent a single body and single speaking voice. Lastly, cutting horizontally resists the encoding process as sutured fragments of magnetic information form new sonic mutations which would be impossible through the vanilla tape recorder design alone.

As was discussed in the playback therapeutic technique, obsessive compulsive thoughts or fears could be weakened through an externalized subvocalization of these thoughts via the tape recorder. Burroughs and subsequently Hayles follow a similar train of thought in proposing the relations of tape-recorder-as-body and body-as-tape-recorder. Burrough's saw language and the subvocalized inner monologue to be a virus and saw the cut-up technique as a means to *'expunge the habitual patterning inscribed by language'*.[38] LaBelle explains this neatly as *'Burroughs perceives the body itself as a prerecorded script, a pre-written 'ticket' to be exploded'* [39] As audio recording and playback mediums became more ubiquitous they created new arrangements or relations between human, voice, and hearing. Hayles describes this disembodied voice as part of a new kind of postmodern subjectivity that is *'fractured by temporal dislocations and riven by disconcerting metatamorphoses, this evolving subjectivity emerges from the instabilities produced when voices are taken out of bodies and bodies find themselves without voices'*[40] When bodies find themselves without voices, as will be seen through Jonathan Sterne's in the next section, there is yet the possibility to build one anew.

Jonathan Sterne's Volutes

Jonathan Sterne is a sound and voice researcher who lost substantial speaking ability after undergoing surgery to remove a thyroid cancer tumor. During the surgery, his recurrent laryngeal nerve was removed which is needed for vocalizing, swallowing, and breathing. In Jonathan's case, this unfortunate circumstance provided him with a new perspective on a subject which he had been researching throughout his entire professional career. His work entitled Diminished Faculties proposes an impairment phenomenology which *'does not assume a subject in command of their own faculties.'*[41] This is defined in opposition to the phenomenological traditions stemming from Husserl to Heidegger and Merleau-Ponty who believed that *'a subject… could produce a transcendental account of consciousness and its*

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[37] Hayles, *How We Became Posthuman*.
[38] LaBelle, "8 Raw Orality: Sound Poetry and Live Bodies."
[39] LaBelle.
[40] Hayles, "Voices Out of Bodies and Bodies Out of Voices."
[41] Sterne, *Diminished Faculties*.

possibilities'.[42] Impairment phenomenology introduces interesting challenges in subjectivity like *'how to account for an experience of self that is unstable and ultimately not fully available?*' and *'how to account for selves, subjects, and experiences while presupposing the constant possibility of impairment, error, breakdown, and incommensurability?'*[43]

To mitigate the loss of communication due to his impaired voice, Sterne experiments with portable audio equipment, which he calls the dork-o-phone, to recover or re-gain speaking abilities. However, he encounters a problem as *'the dork-o-phone's speaker cannot be positioned over the mouth, and yet the mouth is supposed to be the visual and sonic point of origin of the voice'* [44] This situation in which Sterne's new voice box is, in an uncanny way, located outside the body and distant from the mouth resembles Hayles' disembodied voice. Yet, Sterne seeks in the dork-o-phone *'a kind of instrumental relationship, a merger of subject and object, where the articulation of body vocal practice, and device becomes 'just my voice' in a moment of speech or vocalization'*[45] This longing for Sterne reflects a contradictory philosophy of voice which simultaneously employs a Cartesian Dualism separating self (subject) and body/animated-voice (object); A practice-based model in which voice is materially located both historically and culturally through body techniques; And an autopoietic unfolding of voice through self transformation. To explore this contradictory philosophy, Sterne creates an imaginary art exhibit called 'In Search of New Vocalities' which *'consider[s] vocality as a kind of representational field and… the stakes of locating voices on bodies—onto mouths, hands, prosthetics, larynxes—or outside bodies as objects in the world, or as attached to nonhuman objects'*[46]

In the volutes room of this exhibit, Sterne accounts historical artifacts and artworks which treat voice as something that exists in a material form outside of the body. First, he considers Meso-American speech scrolls called volutes which depict voice through sculpture in the form of spiral forms such as seashells, feathers, precious stones, birds, and flowers. These symbolic forms depict speech as something with a unique beauty or signature representative of the speaking subject, yet in the act of speech they untether from the speaker and drift in accordance with the forces of the wind. Next, is a consideration of banderoles as appearing in European paintings, stained glass, and tapestry. Similar to Meso-American volutes, banderoles depict speech as outside of the body emanating from the mouth in the form of words inscribed upon an unspooling scroll. The shapes of these scrolls *'were used to represent subjects' speech but also potentially their emotions or state of mind'* [47] Here, voice is represented through the dominant medium of the epoch, the written word on paper, which suggests uniqueness in signature of the speaker through not only gestures of the hand in writing, but also through the geometrical curvature of the unspooling scroll and its placement amongst the subjects in an artwork. Third is the phonautogram which is the first mechanical invention to inscribe sound created in 1859. Vibrations of the air are transferred from a horn's diaphragm into 2D waveforms

[42] Sterne.
[43] Sterne.
[44] Sterne.
[45] Sterne.
[46] Sterne.
[47] Sterne.

inscribed by a stylus imprinting on paper both up and down and left to right with the automated rhythm of turned-crank. Sterne remarks that the phonautogram signals a shift in treating *'voicing as a unique event, independent of the speaker… defined by a complete absence of any visual depiction of the subject'*[48] Further developments of this analog mechanical model of speech occur at AT&T's Bell Laboratories, as is mentioned earlier via Tukey's Cepstrum, in the form of sound waveforms and spectrograms. These modes of representation are most inclined towards applications of digital computational hearing such as *'for speech recognition, allowing computers to compare a spoken utterance to a library of stored reference patterns for each word or phrase'* This treats voice just *'like any other sound, since a spectrogram or waveform is equally at home representing music, ambient noise, or a street scene'* [49] As this homogenized form of voice representation has become the dominant mode due to its compatibility and integration into the standard designs of computers and other electronic media, Sterne highlights contemporary artists who have searched for other modes of representation. Among them is Louisa Bufardeci's embroidered waveform artworks which extends Zeynep Buluts idea of tactile speech to a telephone conversation separating speakers by varied thread colors. Second is Lozano-Hemmer's Volutes, which inscribe voice into a 'three-dimensional snapshot of sound in motion, or air disturbance in space'[50] While all of these examples depict voice as occurring outside of the body, this history reveals that its representation coincides with the dominant material forms available to creators at the time of conception. The contemporary works signal a human impulse to explore new vocalities which provide alternative modes of engagement through multi-sensory forms of speech.

Conclusion

In tracing the emergence of quantized theories of voice and hearing, their subsequent technological inventions, and manifestations within the field of therapy as well as artforms of musique concrète and the cut-up provided frameworks for treatment of voice within an electronically mediated landscape. In chasing the objective communicative unconscious, therapeutic techniques have reduced the agency of human analysts in favor of automated transcription tools equipped with machine learning models trained to detect symptoms of depression through the materialization of voice as a windowed signal. Adjacent, the analytical value of a patient's voice emphasized small sample times as opposed to the deep states of conversation foundational to the talk-therapy approach. Lastly, the sensory techniques used to interpret patients' voices occurred in closed doors without the patient and have subsequently been passed over to machine hearing tools like Sonde's. These circumstances allude to a reduction in human capacity or willingness to form subjective interpretations of innerlife in favor of objective machine analytic tools. The traditions of granular synthesis and the cut-up technique, which share many common ancestors and timelines, offer alternative relations amongst humans' sense of self and material inscription mediums of voice as a signal. These traditions along with Sterne's phenomenology of impairment emphasize a Do-It-Yourself (DIY) nature of voice. The phenomena of voice and sense of self are in a continual transformation

[48] Sterne.
[49] Sterne.
[50] Sterne.

with regard to technology. It is within each of us the possibility to experiment, discover, and reconstruct our voices anew. This may require an intentional resistance to the operations of voice inscription and playback mediums as in the case of Burroughs. It may require alternative relations between bodily expression and sound as in the case of Xenakis's UPIC. And it may require experimentation with a variety of vocalization technologies as is in the case of Sterne's dork-o-phone. A recognition of voice's disembodiment is not an end, but a beginning in the search for new embodiments.

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